TRIBHUVAN UNIVERSITY

INSTITUTE OF SCIENCE AND TECHNOLOGY

An Internship Report On

"Software Developer Intern"

At

ENDEAVOR NEPAL

Under the Supervision of

Mr. Janak K Lal

Asian College of Higher Studies

Submitted To:

Department of Computer Science and Information Technology

Asian College of Higher Studies

Ekantakuna, Lalitpur

In Partial fulfillment of the requirements For the Bachelor of Science in Computer Science and Information Technology

Submitted By:

Sonish Upadhyaya (26741/077)

June, 2025

MENTOR’S RECOMMENDATION

I hereby recommend that this internship report prepared under my supervision by **Sonish Upadhyaya** entitled SOFTWARE DEVELOPER **INTERN** in partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Information Technology be processed for the evaluation.

…………………………………………

Mr. Suman Bhandari

Software Engineer

Endeavor Nepal PVT. LTD.

SUPERVISOR'S RECOMMENDATION

I hereby recommend that this internship report is prepared under my supervision by **Sonish Upadhyaya** in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science and Information Technology is recommended for final evaluation. In my best knowledge, this is an original work in Computer Science and Information Technology.

....

Mr. Janak K Lal

Supervisor

Department of Computer Science and IT

Asian College of Higher Studies

Ekantakuna, Lalitpur

ASIAN COLLEGE OF HIGHER STUDIES
ACHS
COLLEGE OF IT AND COMPUTER SCIENCE
(Affiliated to Tribhuvan University)

LETTER OF APPROVAL

This is to certify that this internship report prepared by SONISH UPADHAYAY entitled SOFTWARE **DEVELOPER INTERN** in partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Information Technology has been well studied. In our opinion it is satisfactory in terms of scope and quality as a project for the required degree.

-- **SIGNATURE of Supervisor** Mr. Janak K Lal Asian College of Higher Studies Ekantakuna, Lalitpur	-- **SIGNATURE of Mentor** Mr. Suman Bhandari Endeavor Nepal PVT.LTD Sankhamul-31, Kathmandu
--- **SIGNATURE of Internal Examiner**	-- **SIGNATURE of External Examiner**

CERTIFICATE OF COMPLETION

ENDEAVOR NEPAL PVT. LTD.

REGD. NO. 124543/071/072

ENDEAVOR NEPAL

Ref. No.: AMD-158 **Date: 17th June, 2025**

TO WHOM IT MAY CONCERN

Re: Internship Completion Certificate

This is to certify that **Mr. Sonish Upadhyaya** has successfully completed his internship at **Endeavor Nepal Pvt. Ltd.** from **23rd February, 2025** to **24th May, 2025** as an **Intern-Software Developer**.

During the internship period, he consistently demonstrated enthusiasm, professionalism, and a strong willingness to learn. His contributions to the team and assigned tasks were appreciated.

This certificate is issued as a formal confirmation and recognition of the time, efforts, and commitment he exhibited throughout the internship. We believe this experience has provided him with valuable insights and learning opportunities relevant to his academic and professional growth.

We wish him continued success in all his future endeavors.

Sincerely,

ENDEAVOR NEPAL

....

Sunita KC
Managing Director

Tel: +977- 9851122968 | Shankhamul, Kathmandu, Nepal | www.endeavornepal.com | info@endeavornepal.com

ACKNOWLEDGMENT

I want to sincerely thank everyone who helped and mentored me during my internship.

My sincere gratitude goes out to my academic supervisor, Mr. Janak K. Lal, Lecturer in the Department of Computer Science at Asian College of Higher Studies (ACHS), for his unwavering encouragement, support, and wise counsel during the internship. His knowledge, tolerance, and helpful criticism greatly influenced the caliber of my work.

Additionally, I would like to thank Mr. Pranaya Nakarmi, the Faculty Coordinator at ACHS College, for his helpful counsel and helpful demeanor. His encouragement and prompt suggestions were crucial to my internship's successful conclusion.

I want to express my gratitude to Endeavor Nepal Pvt. Ltd. for giving me the chance to join their creative and motivating team. I would especially like to thank Mr. Suman Pandey and Mr. Suman Bhandari for their technical assistance, encouragement, and mentoring during my tenure at the company. Additionally, I would like to thank Mr. Sunil Magar and the entire Endeavor Nepal team for their kind support, cooperation, and help throughout the internship.

Finally, I want to express my gratitude to all of my friends, peers, and anyone else who helped me out in any way during this internship or while I was writing this report. I sincerely thank everyone who helped to make this educational experience fruitful and rewarding.

With sincere appreciation,

Sonish Upadhyaya

ABSTRACT

This report outlines the experience of working as a software developer intern at Endeavor Nepal Pvt. Ltd., a well-known IT firm in Nepal involved in significant government projects, is described in this internship report. Using cutting-edge frameworks like Laravel, Django, and Next.js, the internship offered invaluable practical experience in full-stack web development. Developing and maintaining features for two significant government projects—one with the Teacher Service Commission (TSC) using a Django+ Next.js stack and the other with the Nepal Agricultural Research Council (NARC) using Laravel `were among the main duties. With emphasis on system design, API development, authentication methods, and user interface improvements, the internship provided the chance to put theoretical knowledge from the undergraduate program to practical use. The technical expertise acquired, problem-solving strategies used, and professional growth attained while supporting the digital transformation of Nepal's public sector are highlighted in this report.

Keywords: *Software Development, Laravel, Django, Next.js, Government Projects, Web Application, Full Stack Development, Nepal Agricultural Research Council, Teacher Service Commission*

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LIST OF ABBREVIATION

API	Application Programming Interface
CI/CD	Continuous Integration / Continuous Deployment
CMS	Content Management System
CRUD	Create, Read, Update, Delete
CSRF	Cross-Site Request Forgery
DM	Dry Matter
EDN	Endeavor Nepal Pvt. Ltd.
IDS/IPS	Intrusion Detection / Prevention System
JWT	JSON Web Token
MVC	Model-View-Controller
MVT	Model-View-Template
NARC	Nepal Agricultural Research Council
RBAC	Role-Based Access Control
R&D	Research and Development
STRO	Survey Department Teacher Record Online
TSC	Teacher Service Commission
UNIC	United Nations Information Centre
VMIS	Vehicle Management Information System
WRRDC	Water Resources Research and Development Centre

CHAPTER 1: INTRODUCTION

1.1 Introduction

The Bachelor of Science in Computer Science and Information Technology program under Tribhuvan University mandates students to complete a professional internship worth six academic credits, requiring a minimum duration of ten weeks (180hours). This internship is structured to offer real world exposure, allowing students to apply theoretical knowledge in practical settings within professional environments.

The internship was completed at the renowned IT firm Endeavor Nepal Pvt. Ltd. (EDN), which is situated in Sankhamul-31, Kathmandu. Since its founding in 2014, EDN has made a name for itself in Nepal as a reliable supplier of complete ICT solutions to the public and private sectors. Network and server infrastructure, cloud services, virtualization, ICT consulting, hardware solutions, and software development are among the company's areas of expertise. Underpinned by its core values of excellence, trust, proactiveness, transparency, and teamwork, EDN prioritizes long-term client relationships, technological innovation, and service reliability with a team of seasoned IT professionals and business experts.

Contributing to significant government digitalization initiatives, particularly with the Teacher Service Commission (TSC) and the Nepal Agricultural Research Council (NARC), was part of the job description for a software developer intern. The main components of the technical stack were Next.js, Django, and Laravel, all of which were combined into a microservices architecture. Debugging code and implementing essential modules like search functionality, dynamic link management, login page password policies, CAPTCHA validation, and password reset/update flows were among the main contributions. Through the integration of Celery, Celery Beat, and Flower, the internship also provided experience in background task processing and monitoring.

1.2 Problem Statement

Businesses and organizations are working to implement technology-driven solutions in Nepal's quickly changing digital landscape in order to increase productivity, accessibility, and user engagement. Nevertheless, there are still few software programs designed specifically for Nepal's distinct linguistic, cultural, and infrastructure context. A large number of current systems are either imported, excessively generic, or ineffective at meeting the needs of localized users. Furthermore, developers frequently encounter obstacles like restricted access to organized datasets, inadequate documentation for support of local languages, and a lack of hands-on experience with actual software development procedures.

There is a vital chance for an intern software developer at Endeavor Nepal to help create user-centric, localized, and scalable software systems that close the knowledge gap between theory and practice. This calls for concentrating on creating reliable frontend and backend systems, incorporating intelligent automation where appropriate, and making sure the finished product satisfies the requirements of Nepali organizations and users.

1.3 Objectives

The main objectives of the internship are:

- To implement theoretical knowledge of software development into real-world projects, enhancing problem-solving, teamwork, and communication skills.
- To gain hands on experience in full stack development, including frontend interfaces, backend logic, and database integration.
- To collaborate in an agile development environment and contribute to building scalable and user-friendly software solutions.
- To participate in code reviews, testing, and debugging processes to improve software quality and maintainability.

1.4 Scope and Limitation

1.4.1 Scope

The scope of this internship is as follows:

- Strengthening core programming skills in PHP, Python, and JavaScript to support scalable application development.
- Gaining practical experience in full-stack development using Laravel for backend services, Django for data processing and APIs, and Next.js for dynamic frontend interfaces.

- Building and integrating RESTful APIs across frameworks to ensure smooth communication between frontend and backend components.
- Developing secure and user-friendly web applications by implementing authentication, authorization, and form validation features.
- Collaborating in an agile workflow, contributing to sprint tasks, and learning version control best practices using Git.

1.4.2 Limitations

Despite the broad scope, the internship also had certain limitations:

- The 10–12-week duration limited the exploration of advanced topics such as microservices, advanced state management in React/Next.js, and asynchronous task queues.
- Deployment experience was minimal, with limited hands-on exposure to containerization (Docker), CI/CD pipelines, or cloud platforms like AWS or Azure.
- Integration between Laravel and frontend frameworks was mostly backend-focused, limiting deep involvement in UI/UX decisions or advanced frontend interactivity.
- Cross-framework project coordination presented challenges in managing shared data models and consistent API design across Laravel, Django, and Next.js.

1.5 Report Organization

Table 1.1 Report Organization

Chapter	**Title**	**Description**
Chapter 1	**Introduction**	This chapter provides an overview of the internship program, including its objectives, relevance to academic and professional development, and overall scope and limitations. It introduces the domain of software development and sets the context for the technical work carried out.

Chapter 2	**Organization Details and Literature Review**	This chapter presents the organizational background, team structure, and the technology stack used during the internship. It includes a brief overview of Laravel, Django, and Next.js frameworks and their roles in the assigned projects.
Chapter 3	**Internship Activities**	This chapter describes the specific tasks and responsibilities undertaken during the internship. It includes developing and debugging web applications, API integration, backend logic implementation, database operations, and collaboration in agile workflows.
Chapter 4	**Conclusion and Learning Outcomes**	The final chapter summarizes the overall internship experience, technical and soft skills gained, challenges faced, and contributions made. It reflects on the professional growth achieved and outlines recommendations for future learning in software development.

CHAPTER 2: ORGANIZATIONAL DETAILS AND LITERATURE REVIEW

2.1 Introduction to Organization

Endeavor Nepal Pvt. Ltd. is a prominent IT company based in Nepal, established in 2014. The company specializes in providing comprehensive ICT solutions, including web and mobile application development, tailored to various industries. With a focus on innovation and quality, Endeavor Nepal has been instrumental in delivering customized software solutions that cater to the unique needs of its clients.

Key Achievements

- Provided Annual Maintenance Contract (AMC), firewall systems, antivirus protection, and a Digital Library System to the United Nations Information Centre (UNIC), Kathmandu.
- Implemented firewall setup and antivirus protection for the Water Resources Research and Development Centre (WRRDC).
- Designed and deployed the complete system software with advanced security features for the Government of Nepal Ministry of Federal Affairs and General Administration Department of Personal Records (Teacher)(https://stro.gov.np).
- Developed, deployed, and secured the Vehicle Management Information System (VMIS) for the Department of Roads, Government of Nepal (https://vmis.dor.gov.np/system/login).
- Created the full frontend, content management system (CMS), API, database, and deployed secure infrastructure for the Nepal Agricultural Research Council (https://narc.gov.np/).

Collaborations

Endeavor Nepal has collaborated with several key governmental and international institutions, offering software development, IT security, and network management solutions tailored to their specific operational needs. These collaborations showcase the company's technical capacity and reliability in critical systems development and support.

Organization Details

Table 2.1 Organization Details

Organization Name	Endeavor Nepal
Established Date	2014 AD
Organization Type	Private
Address	Sakhamul-31, Kathmandu, Nepal
Contact Number	+977 9851133968
Email	Info@endeavornepal.com
Website	https://www.endeavornepal.com

Figure 2.1 Logo of Organization

2.2 Organizational Hierarchy

Endeavor Nepal follows the following diagram. Major positions include:

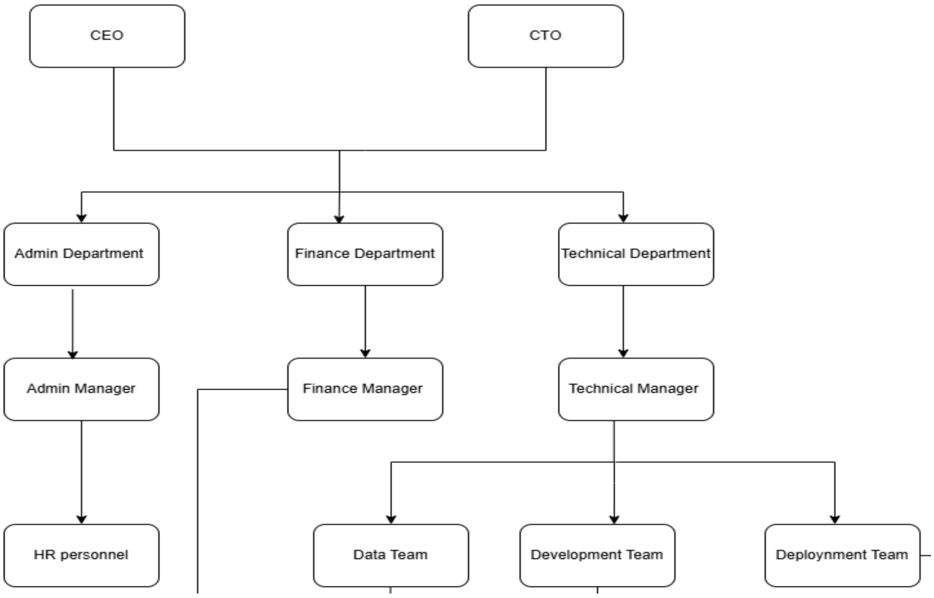

Figure 2.2 Organizational Hierarchy of Endeavor Nepal

2.3 Working Domains of Organization

The working domains of Endeavor Nepal span across software development, network security, enterprise infrastructure, and government system automation. The organization leverages modern technology stacks and international affiliations to deliver scalable and secure solutions.

Software Development and System Integration

Endeavor Nepal specializes in building full-stack web applications for government and enterprise clients. The company delivers frontend and backend systems, CMS APIs, relational databases, deployment pipelines, and secure hosting infrastructure. Their expertise includes both customized platforms and government-scale systems such as VMIS (Vehicle Management Information System) and STRO (Survey Department).

Cybersecurity and Firewall Solutions

With affiliations to Fortinet, SOPHOS, and Juniper Networks, Endeavor Nepal provides robust cybersecurity solutions. This includes the setup of enterprise-grade firewalls, antivirus systems, intrusion detection/prevention systems (IDS/IPS), and endpoint protection for organizations like the United Nations Information Centre and WRRDC.

Cloud and Enterprise IT Infrastructure

The company provides infrastructure design and implementation services using technologies from partners such as Hewlett Packard Enterprise, Dell EMC, QNAP, and Microsoft 365. Solutions include server deployment, virtualization, data backup systems, and cloud service integration.

E-Governance and Public Service Automation

Endeavor Nepal actively works with government bodies to develop and secure software platforms that digitize public service operations. These solutions streamline workflows, enhance data accessibility, and provide real-time monitoring and reporting tools for public administrators.

2.4 Description of Intern Department

The internship as a Full Stack Developer Intern at Endeavor Nepal Pvt. Ltd. officially began on February 23, 2025, and continued until completion as part of the academic requirement. The intern was placed within a compact yet highly experienced Development Department consisting of 12 members, including 3 System Administrators, 4 Full Stack Developers, and 1 Intern Developer.

The department operated under the direct supervision of Mr. Suman Bhandari, who provided consistent technical guidance and mentorship throughout the internship. The team was responsible for a variety of end-to-end software development projects, primarily using the Laravel framework, while also involving technologies like Django and Next.js depending on the project scope.

Projects within the department typically involved a full spectrum of development tasks—from frontend design and API development to database structuring, deployment, and system-level security implementations. A strong emphasis was placed on writing modular and scalable code while maintaining adherence to clean architecture and coding standards.

The department fostered a collaborative and goal-oriented culture, promoting hands-on learning and agile practices. This environment provided the intern with direct exposure to the realities of professional software development. Regular involvement in activities such as coding, debugging, client requirement analysis, and live system deployment contributed significantly to strengthening technical capabilities and building a deeper understanding of modern full-stack web application development.

Table 2.2 Description of Intern Department

Start date	February 23, 2025
End Date	August 23, 2025
Total Duration	24 weeks
Office Hour	10:00 am – 6:00pm
Working Days	6 days a week (Sunday-Friday)
Position	Software developer Intern
Mentor	Mr. Suman Bhandari

2.5 Literature Review

2.5.1. Secure and Scalable Role-Based Access Control in Web Applications

One well-liked method of controlling user access in a web application according to their roles is Role-Based Access Control, or RBAC. Consider it similar to giving different keys to different people: an administrator may have the key to everything, while a normal user can only access the most basic features. By ensuring that only authorized users can view or alter sensitive information, such as admin dashboards or content management panels, this method helps maintain the app's security. Additionally, it facilitates adherence to corporate policies and eases user management. Working with RBAC with tools like Django, NextAuth.js, or Laravel while learning web development—especially during internships—teaches you key concepts like handling user sessions or tokens, controlling access behind the scenes (middleware), and displaying or hiding portions of the user interface based on who is logged in.

Strong security measures like guarding against cross-site request forgery (CSRF), utilizing safe cookies with JWT (a method to keep users logged in securely), and facilitating easy, safe login procedures are also included in a good RBAC system. When implemented properly, RBAC offers users seamless and secure experience regardless of the system's user base, while also making it easier to expand and maintain your app.

2.5.2. Asynchronous Task Queues with Celery for Background Processing

In scalable web applications where real-time response is crucial, background task processing is crucial. When combined with Django, tools like Celery allow long-running tasks like database cleanup, report generation, and email and SMS notifications to be completed asynchronously. Celery queues tasks using message brokers like Redis or RabbitMQ and collaborates with Celery Beat for recurring scheduling. Using Celery in conjunction with Flower, a monitoring tool, in real-world internship settings guarantees that queued jobs are reliably completed and offers real-time task visibility. Applications can reduce server load and maintain responsiveness by assigning heavy tasks to background workers. Building contemporary, production-ready backend systems requires an understanding of asynchronous design patterns.

2.5.3. Modern CMS Design for Dynamic Web Content Delivery

Non-technical users can update, arrange, and publish dynamic content across websites using a Content Management System (CMS), which acts as a backend interface. Developers can create modular CMS platforms with role-based access, database integration, and dynamic page rendering by utilizing frameworks such as Laravel, Django, and Next.js. Creating CMS platforms during internships enhances knowledge of routing, form handling, MVC/MVT architecture, and CRUD operations (Holovaty, 2022). Complex backend logic is abstracted from end users by well-structured CMS systems, enabling content updates (such as announcements, rental details, and course listings) without requiring direct code changes. Additionally, developing CMS systems improves one's proficiency in data validation, secure input handling, and API developmental which are essential for enterprise applications.

2.5.4. Integrating Frontend Templates with Backend APIs in Full-Stack Development

Backend APIs and frontend elements (HTML, CSS, and JavaScript) must integrate seamlessly into modern full-stack development. Django template directives, Blade (Laravel), or Next.js pages are frequently used by interns working on real-world systems, such as rental platforms or school websites, to render dynamic content from backend services. Consuming REST APIs, securely processing form submissions, controlling session state, and preserving a consistent user experience across routes are all part of this process. Creating this link between the frontend and backend offers knowledge about middleware pipelines, secure data rendering techniques, and HTTP methods. Additionally, it allows developers, designers, and content managers to collaborate while maintaining a clear division of responsibilities.

CHAPTER 3: INTERNSHIP ACTIVITIES

3.1 Roles and Responsibilities

3.1.1 Roles

Full Stack Developer Intern

- Laravel Developer
- Django + Next.js Developer
- Module Tester and Debugger
- R&D Contributor for Feature Enhancements

3.1.2 Responsibilities

- Develop code for various project modules, ensuring functionality, security, and scalability across backend and frontend components.
- Design and implement RESTful APIs to handle data storage and retrieval from the database, supporting dynamic application features.
- Integrate API data into frontend interfaces using Next.js and Django, ensuring responsive and interactive user experiences.
- Collaborate with the team to gather and analyze client requirements, translating them into technical tasks and actionable development goals.
- Conduct module testing and debugging to identify, isolate, and resolve software issues prior to deployment.
- Provide live demonstrations and progress updates to clients, showcasing completed features and gathering feedback.
- Fix identified bugs during internal QA or client review to ensure stable, production-ready deployments.

3.2 Weekly Log

The following weekly logs provide a detailed account of the technical activities carried out during the internship. The logs serve as a chronological record of week-by-week experiences during the internship, covering encountered challenges, employed problem-solving strategies, and achieved milestones.

Table 3.1 Task Assigned

Task Assigned	Description
Module Development	Designed and implemented various backend modules using Laravel and Django frameworks. Responsibilities included writing clean, scalable, and maintainable code based on system requirements and project specifications.
API Development	Created RESTful APIs to perform CRUD operations, enabling data exchange between the backend and frontend. Ensured secure and efficient communication between different parts of the application.
Frontend Integration	Integrated API data into the frontend using Next.js. Ensured smooth user interface interaction with real-time data and maintained responsiveness across devices.

Table 3.2 Internship Workflow

Week	Planned Activities
Week 1 (23rd February-1st March)	• Introduced to the organization, its objectives, and technology stacks, particularly the Laravel framework. • Assigned a basic full-stack CRUD project to understand Laravel's MVC structure. • Set up the local development environment, including server configuration and database connectivity. • Studied the codebase architecture of existing projects to understand backend logic and frontend API integration. • Oriented with the development team's workflow, version control practices via GitLab, and coding standards.

Week 2 (2nd March- 8th March)	• Developed a global search feature in the frontend, enabling real-time database result display based on user input.
	• Designed and implemented the Link Management module in the CMS:
	    o Created database models and migrations.
	    o Developed backend logic using Laravel's controller-service-repository pattern.
	    o Built frontend UI using Laravel Blade templates with CRUD functionality.
	• Created an API endpoint to serve link data dynamically to the frontend.
	• Conducted feature testing and followed Git best practices for code commits and merging to the dev branch.

Week 3 (10th March- 18th March)	• Assigned the development of the Staff Management module within the CMS:
	    o Designed models for staff and staff types (designations) and performed necessary migrations.
	    o Developed backend logic and Laravel Blade-based CRUD UI.
	    o Integrated data filtering based on designations and date.
	• Developed API endpoints and linked them with the frontend.
	• Performed debugging and testing of implemented modules and applied necessary bug fixes.

Week 4 (19th March- 25th March)	• Focused on strengthening application security and authentication features:
	    o Enforced password policy through custom validation logic in Laravel.
	    o Implemented password reset and recovery via email with secure token generation.

	◦ Configured email settings and designed Laravel Blade templates for reset instructions. • Integrated CAPTCHA protection using the mews/captcha package to safeguard login flows from automated attacks.
Week 5 (27th March- 3rd April)	• Conducted technical R&D and delivered a presentation justifying the adoption of Next.js and Django for the upcoming TSC system. • Compared modern frontend/backend stacks based on scalability, security, and rendering capabilities. • Performed whitebox testing on the NARC project: ◦ Reviewed code logic and tested modules for potential issues. ◦ Resolved identified bugs to improve production stability.
Week 6 (4th April- 9th April)	• Initiated development for the TSC system: ◦ Analyzed project documentation to define system objectives, modules, and user roles. ◦ Outlined database schema and established table relationships for backend development. • Began Next.js frontend setup: ◦ Developed reusable UI components including forms, navigation bar, and footer. ◦ Followed modular design practices and styling conventions to ensure consistency and maintainability.

Week 7 (10th April- 16th April)	• Researched and implemented asynchronous task handling in Django using Celery, Celery Beat, and Flower: • Integrated Celery for managing background job queues (e.g., email sending). • Set up Flower for real-time task monitoring. • Configured Celery Beat with cron jobs to automate scheduled tasks. • Applied the setup to auto-generate job application reports at fixed intervals.

Week 8
(18th April-
25th April)
	• Oversaw post-deployment support for the live NARC system after official launch on 20th Baishakh:
	    o Assisted client in identifying and resolving real-time issues and system bugs.
	    o Monitored system performance and implemented prompt fixes to maintain stability.
	• Gained exposure to production-level debugging, error tracking, and client-side communication.

Week 9
(27th April-
2nd May)
	• Continued post-launch support for the NARC system, closely collaborating with the client:
	• Actively responded to feedback and new feature requests.
	• Implemented necessary updates based on client suggestions to ensure smooth functionality and user satisfaction.
	• Worked on the TSC project, specifically on the Voucher Detail Module:
	• Developed a Django API endpoint that:
	    o Accepts user input.
	    o Sends the request to the FCGO API to fetch voucher-related data.
	    o Generates a QR code from the response.

	- Saves the voucher details and QR code into the voucher_table. - Created the Voucher model in Django, drawing inspiration from the legacy Laravel implementation to maintain consistency with previous systems. - Built a static "Voucher Admit Card" page in the Next.js CMS, then: - Connected it to the Django backend by consuming the newly developed API. - Made the admit card dynamic, ensuring it displays real-time voucher information based on backend responses.
Week 10 (3^{rd}May-10^{th}May)	- Continued providing support for the live NATIC system: - Responded to ongoing user feedback and technical queries. - Ensure system reliability by resolving minor issues as they arose. - Focused development efforts on the TSC project, primarily in the Next.js frontend: - Implemented dynamic admin and candidate profile pages, integrating real-time data fetched from the Django backend. - Resolved login authorization conflicts between admin and user accounts, ensuring secure and role-specific access flows. - Mapped API endpoints for editing both admin and candidate profiles to enable seamless data updating from the frontend. - Tested API endpoints for various modules to verify data accuracy and ensure proper mapping with frontend components. - Generated a detailed report based on test outcomes to support future debugging and refinement efforts.

Week 11
(12th May-
17th May)
	• Continued technical support for the NATIC system, addressing minor issues and assisting users as needed.
	• Initiated requirement analysis for a new project:
	    o Conducted research and development (R&D) focused on feed formulation and Dry Matter (DM) calculation.
	    o Designed a flowchart document to visually represent the DM calculation logic and workflow.
	• Progressed with feature development in the TSC project:
	    o Frontend (Next.js):
	        ▪ Developed a universal popup modal component for edit forms and integrated it across all modules to enhance UX consistency.
	        ▪ Implemented filter functionality for each module's listing page:
	            • Built a reusable filter component.
	            • Created a configurable setup to define filter elements per module.
	            • Applied and tested the filters across relevant modules.
	    o Backend (Django):
	        ▪ Developed a new API endpoint for Exam Fee management:
	            • Created the ExamFee model and implemented logic using the project's service-repository architecture.
	            • Validated the functionality using Postman.
	            • Successfully merged the feature into the dev branch after thorough testing.

Week 12 (19th May- 24th May)	- Ongoing support for the NATIC system: - Resolved issues with an external API from NARC that was not displaying correctly on the front end. - Updated the gallery feature to support audio and video files as per the client's new requirements. - Development progress on the TSC project: - Implemented pagination across all listing pages (list.jsx) in the Next.js frontend: - Built a reusable pagination component. - Integrated it consistently into each module's listing view. - Tested API mappings between the frontend and backend to ensure accurate query responses. - Established a clear naming convention for Django API endpoints to prevent confusion during frontend-backend integration. - Initiated backend development for the new project, "Low-Cost Feed Formulation": - Set up the project using Django's service-repository architecture. - Configured essential settings, including database connection and .env variables. - Developed and tested a user registration API endpoint using Postman to validate functionality.

3.3 Description of the Project Involved During Internship

During the internship at Endeavor Nepal Pvt. Ltd., active involvement was centered around two major government-driven software initiatives: the Nepal Agricultural Research Council (NARC) system and the Teacher Service Commission (TSC) portal. Both projects utilized modern full-stack technologies including Laravel, Django, and Next.js, with a strong emphasis on security, scalability, and clean architectural practices.

The NARC system, developed primarily in Laravel, involved key features such as link management, staff management, dynamic search interfaces, and authentication mechanisms. Tasks included designing RESTful APIs, implementing CAPTCHA-based login protection, and enabling secure password recovery through email. Post-deployment support continued after the system's launch, where client feedback was addressed promptly. Feature enhancements were also made, including updates to the gallery module to support audio and video content based on new requirements.

Following the NARC deployment, focus shifted to the TSC portal, which adopted Django for backend development and Next.js for the frontend. Project work began with setting up the architecture, defining database schemas, and implementing modular UI components. Several frontend enhancements were made, including the integration of pagination, filter systems, and popup modals for consistent editing interfaces across modules.

On the backend, a service-repository pattern was followed to ensure maintainability and separation of concerns. API endpoints were developed for modules such as Exam Fee Management, along with the creation of appropriate models and business logic. Asynchronous processing tools such as Celery, Celery Beat, and Flower were configured to automate background tasks like scheduled report generation.

To support seamless frontend-backend communication, a consistent API naming convention was introduced to reduce mapping errors. Extensive testing was conducted using tools like Postman to validate endpoint functionality, with detailed test reports generated to guide further development.

Additionally, groundwork was laid for a new system focused on low-cost feed formulation, where backend setup was completed, including environment configuration and database initialization. Initial endpoints, such as user registration, were developed and tested following modern backend practices.

Overall, the internship provided comprehensive exposure to professional software development processes, with hands-on experience across full-stack development, API design, system architecture, testing, and real-time client interaction.

3.4 Tasks / Activities Performed

Throughout the internship at Endeavor Nepal Pvt. Ltd., involvement spanned a wide range of software development tasks, covering backend logic, frontend interfaces, and systems integration. The experience offered hands-on exposure to real-world development processes, from building core features to ensuring system stability in production environments. Key responsibilities included the following:

1. Backend Module Development

Multiple backend modules were developed using Laravel and Django, following structured design patterns like MVC and MVT, along with a service-repository architecture to maintain modularity. Key modules included:

- Link Management: Designed models, created database migrations, and implemented CRUD functionality.
- Staff Management: Built models for staff and designation types, integrated filters, and connected logic to the frontend.
- Exam Fee Management: Developed the model, service, and view layers to support API-driven interactions.
- Voucher Detail Module: Integrated third-party FCGO API, generated QR codes from response data, and persisted it in the database.

Initial setup for a new feed formulation system was also completed, including environment configuration, database setup, and user registration functionality.

2. RESTful API Creation and Integration

A major part of development involved designing and implementing RESTful APIs for various modules. These endpoints supported secure CRUD operations and enabled seamless communication between frontend and backend systems. Highlights included:

- Building endpoints for staff, link, voucher, profile, and exam fee modules.
- Following consistent naming conventions to reduce confusion during frontend mapping.

- Validating endpoints using Postman, with test reports generated to track query accuracy and reliability.

3. Frontend Integration with Next.js

Frontend responsibilities focused on building a responsive, dynamic user interface using Next.js (Next.js, n.d.). Key tasks included:

- Creating dynamic admin and candidate profile pages.
- Implementing pagination and filtering across module listing pages using reusable components.
- Developing a universal modal component for all edit forms to standardize user experience.
- Integrating APIs with UI components for real-time data rendering and updates.

Frontend modules were consistently styled and structured to ensure maintainability and scalability across the platform.

4. Authentication and Security Enhancements

Security was a top priority, especially for systems handling sensitive user data. Enhancements included:

- Password policy enforcement to encourage strong user credentials.
- Tokenized password reset workflows via email.
- CAPTCHA integration to protect login forms from bot attacks. These features collectively strengthened authentication flows and reduced potential vulnerabilities.

5. Background Processing and Automation

In the TSC portal, asynchronous task management was implemented using Celery, Celery Beat, and Flower:

- Automated periodic tasks such as job application report generation.
- Configured real-time monitoring for task queues and cron jobs.
- Improved backend efficiency by reducing load during high-traffic operations.

This setup allowed for better system performance and ensured reliability under production-level usage.

6. System Testing and Quality Assurance

Quality assurance played a continuous role throughout the internship:

- API endpoints were rigorously tested using Postman for expected query results.
- Integrated modules were verified on the frontend to ensure correct data mapping and error-free rendering.
- Test reports were generated regularly to document functionality, validate edge cases, and support team collaboration.

7. Deployment Preparation and Live System Support

Post-deployment support for the NARC and NATIC systems included:

- Real-time debugging and bug fixing based on user and client feedback.
- Updating features such as the media gallery for audio/video support as per client requirements.
- Monitoring external APIs and resolving issues to ensure consistent performance in the production environment.

Close coordination with clients helped fine-tune system behavior and improve overall user satisfaction.

3.4.1 Implementation Details

The internship focused on the development and enhancement of three key projects: NARC, TSC, and a new Low-Cost Feed Formulation system each with unique technical requirements, technologies, and challenges (Len Bass, 2003). Implementation efforts emphasized building secure, scalable, and maintainable systems aligned with client goals.

NARC Project

The Nepal Agricultural Research Council (NARC) system was developed using the Laravel framework. Implementation involved backend module design, including link management, staff CRUD operations, and a global search feature. These were integrated into the frontend via Laravel Blade templates. Security features such as CAPTCHA and email-based password reset enhanced authentication flows. Post-deployment, ongoing support included bug fixes and feature updates like enhanced audio and video gallery capabilities based on client feedback.

TSC Project

The Teacher Service Commission (TSC) portal utilized Django for backend development and Next.js for the frontend. Work began with project set up defining database schemas, user roles, and UI component libraries. Backend services were created for user registration, job applications, voucher management, and exam fee processing, following a clean service-repository pattern.

The frontend featured reusable components such as pagination, filtering, and modal forms to improve usability. A notable achievement was implementing asynchronous background task processing using Celery, Celery Beat, and Flower (Fowler, 2004), which automated scheduled reports and improved system responsiveness. API endpoints were clearly named and tested extensively to facilitate smooth frontend-backend integration.

Low-Cost Feed Formulation Project

In addition to the above, initial work began on a Low-Cost Feed Formulation system designed to assist with agricultural feed management. The backend was built with Django using the service-repository architecture, with foundational setup completed—including database configuration and environment variables. Key early deliverables included creating a user registration API, tested thoroughly with Postman (Postman, n.d.). Research and documentation efforts also included designing a flowchart for Dry Matter (DM) calculation, laying the groundwork for future development.

3.4.1.1 Development Methodology

The projects followed an Agile and Iterative development methodology, which emphasized continuous improvement, collaboration, and flexibility (Martin, 2008). Work was organized into weekly sprints, with clearly defined tasks ranging from feature development and testing to internal code reviews and client demonstrations.

All code was managed using GitLab, where version control practices such as branching, frequent commits, and pull requests were consistently applied. Mentors regularly reviewed submitted code, providing valuable feedback that supported learning and quality assurance.

Each feature or module underwent a structured development cycle—starting with requirement analysis and planning, followed by implementation, and finally testing and debugging. Techniques like whitebox testing were used to verify internal logic, while peer code reviews and live client feedback sessions ensured that the software evolved in line with real-world expectations.

This iterative process also allowed the team to quickly adapt to changes, including last-minute feedback from clients or stakeholders. The collaborative nature of the methodology ensured that contributions were meaningful and aligned with project goals throughout each phase of development.

3.4.2 Implementation Tools

- Visual Studio Code: Used as the primary code editor for developing and debugging Laravel (Laravel, n.d.), Django, and Next.js projects. Its plugin ecosystem enhanced productivity and code linting.
- GitLab: Utilized for version control and collaborative codebase management. The intern worked with feature branches, rebasing, and merge requests to follow standard CI practices.
- Figma (View Only): Although not deeply involved in UI/UX design, the intern referred to pre-designed mockups in Figma for frontend implementation consistency.
- Postman: Used for API testing, including JWT-based authentication, form submission, and endpoint validations.
- Celery, Celery Beat, and Flower: Integrated for background task scheduling and monitoring within Django-based systems.
- PHPMyAdmin / MySQL Workbench: Tools used to visualize and manage relational database schemas and query execution.

CHAPTER 4: CONCLUSION AND LEARNING OUTCOMES

4.1 Conclusion

The internship at Endeavor Nepal Pvt. Ltd. has been a highly valuable and transformative experience, both technically and professionally. Over the course of the program, extensive involvement in real-world projects allowed for the practical application of academic knowledge while working on scalable, secure, and maintainable web applications using Laravel, Django, and Next.js.

Throughout the internship, direct contributions were made to several critical features, including the development of a role-based content management system, secure JWT-based authentication with CSRF protection, and a background task handling system using Celery, Celery Beat, and Flower. These efforts supported large-scale platforms such as the NARC and TSC systems, both serving significant governmental needs.

The collaborative environment at Endeavor Nepal, combined with exposure to Agile methodologies, Scrum practices, and GitLab version control, fostered strong team communication skills and a deeper understanding of software development lifecycles. Regular feedback, client interactions, and iterative improvements reinforced the importance of planning, testing, and adaptability in professional software projects.

By the conclusion of the internship, a solid foundation was built in full-stack development, along with strengthened problem-solving abilities and a confident approach to handling both backend systems and deployment-level tasks. This experience has played a crucial role in shaping readiness for future roles in the tech industry and laid the groundwork for continued growth as a software engineer.

4.2 Learning Outcomes

- During the internship at Endeavor Nepal Pvt. Ltd., the following key learning outcomes were achieved:
- Acquired hands-on experience in full-stack web development using Laravel, Django, and Next.js, with a strong focus on creating modular, maintainable, and scalable application architectures.
- Implemented role-based authentication systems and developed secure user management workflows using JWT tokens, HttpOnly cookies, and CSRF protection mechanisms in Django-based applications.
- Gained proficiency in handling asynchronous background tasks by configuring Celery, Celery Beat, and Flower for operations such as scheduled email and SMS notifications.
- Strengthened skills in relational and non-relational database management, particularly with MySQL and MongoDB, and improved fluency in integrating frontend and backend through RESTful APIs.
- Expanded frontend development knowledge by integrating dynamic HTML, CSS, and JavaScript within template-based systems, contributing to responsive and user-friendly interfaces.
- Applied Agile development practices, including participation in Scrum-based workflows, daily standups, sprint planning, and regular collaboration with team members.
- Gained a deeper understanding of the importance of writing secure, maintainable, and well-documented code in production environments.
- Built confidence in debugging, testing, and deploying web applications in a collaborative and version-controlled development setting using GitLab.
- These outcomes collectively contributed to both technical growth and professional development, equipping the intern with practical skills and real-world experience in software engineering.

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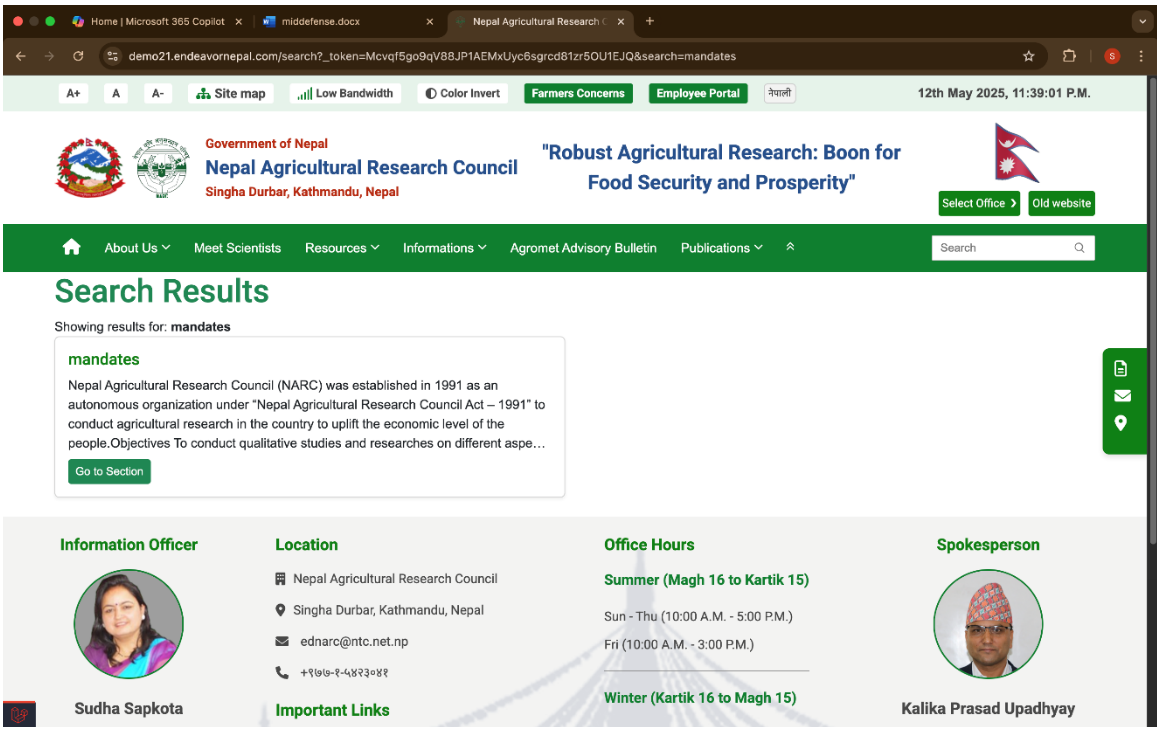
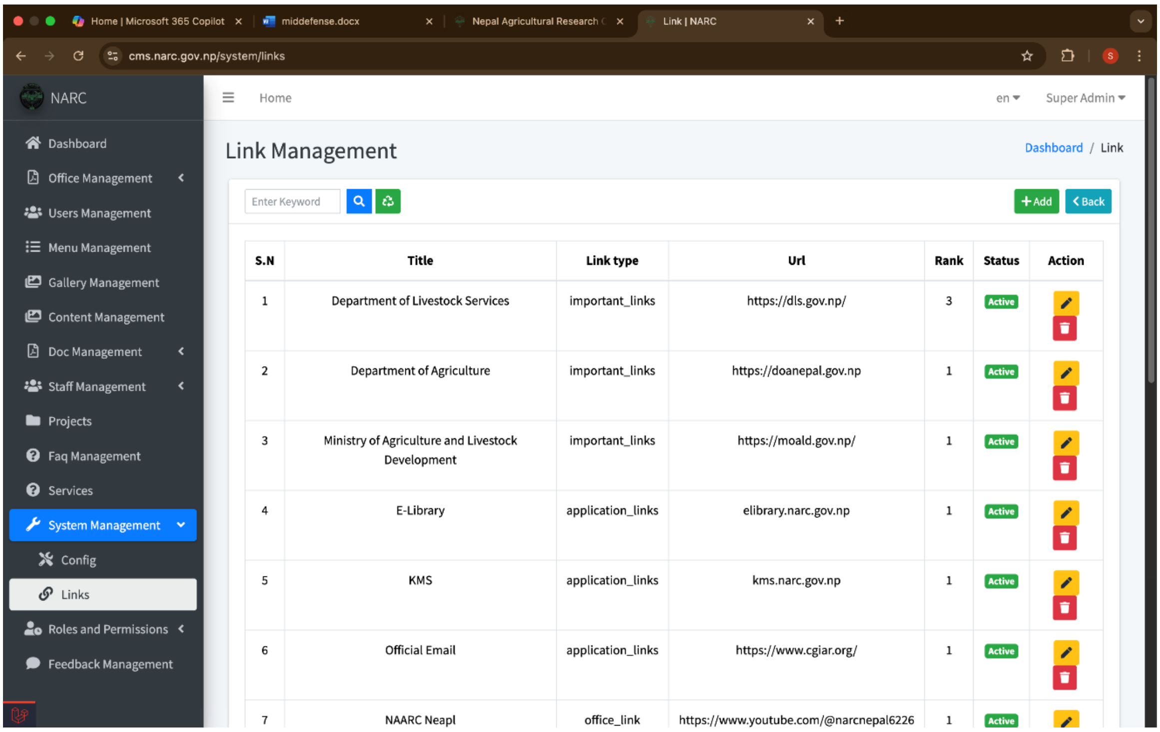
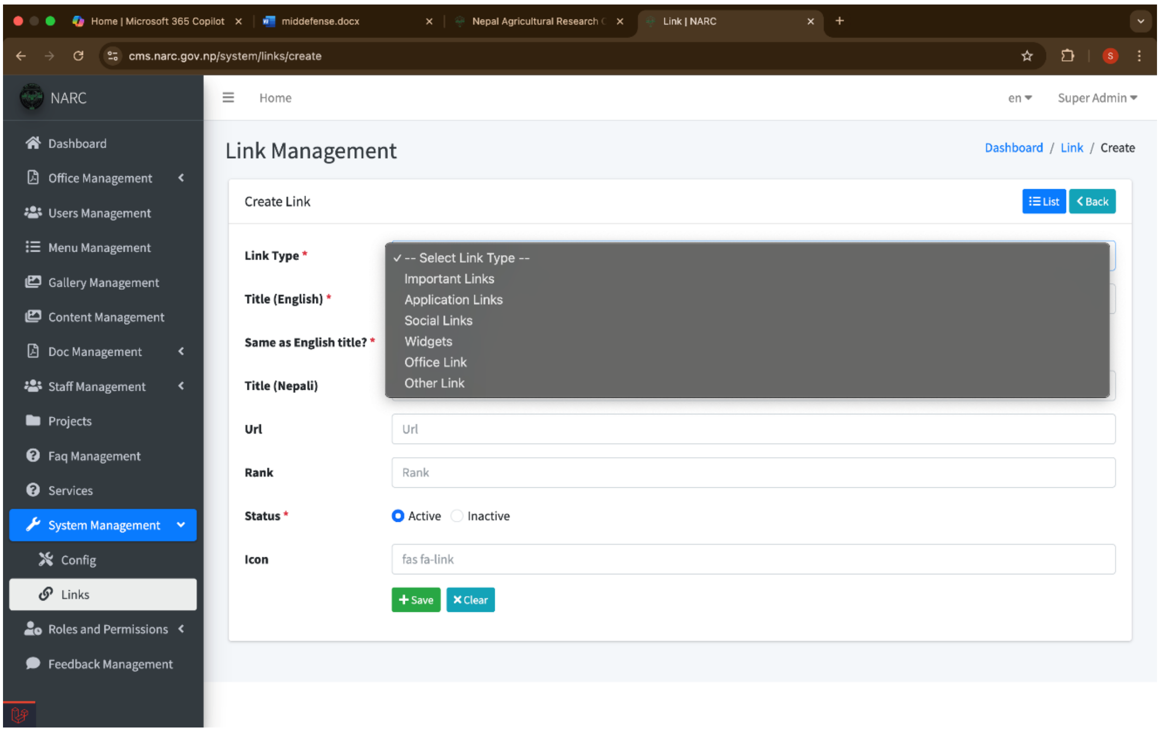
APPENDICES



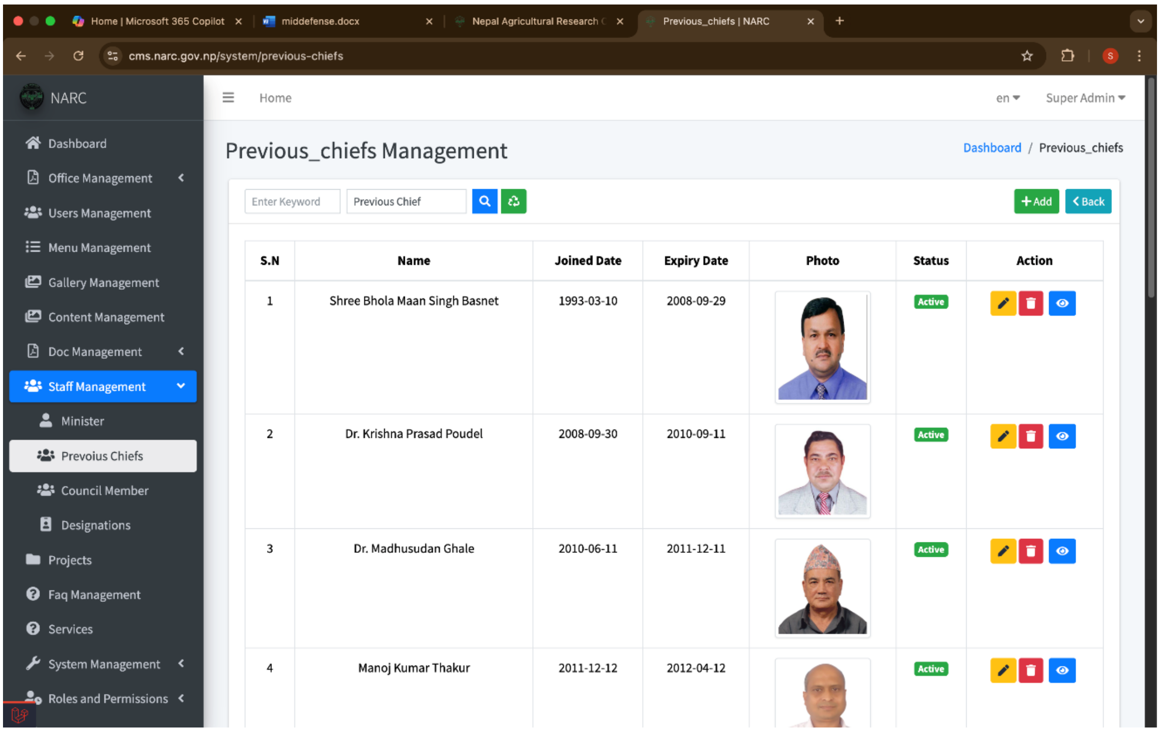
Search Functionality at NARC



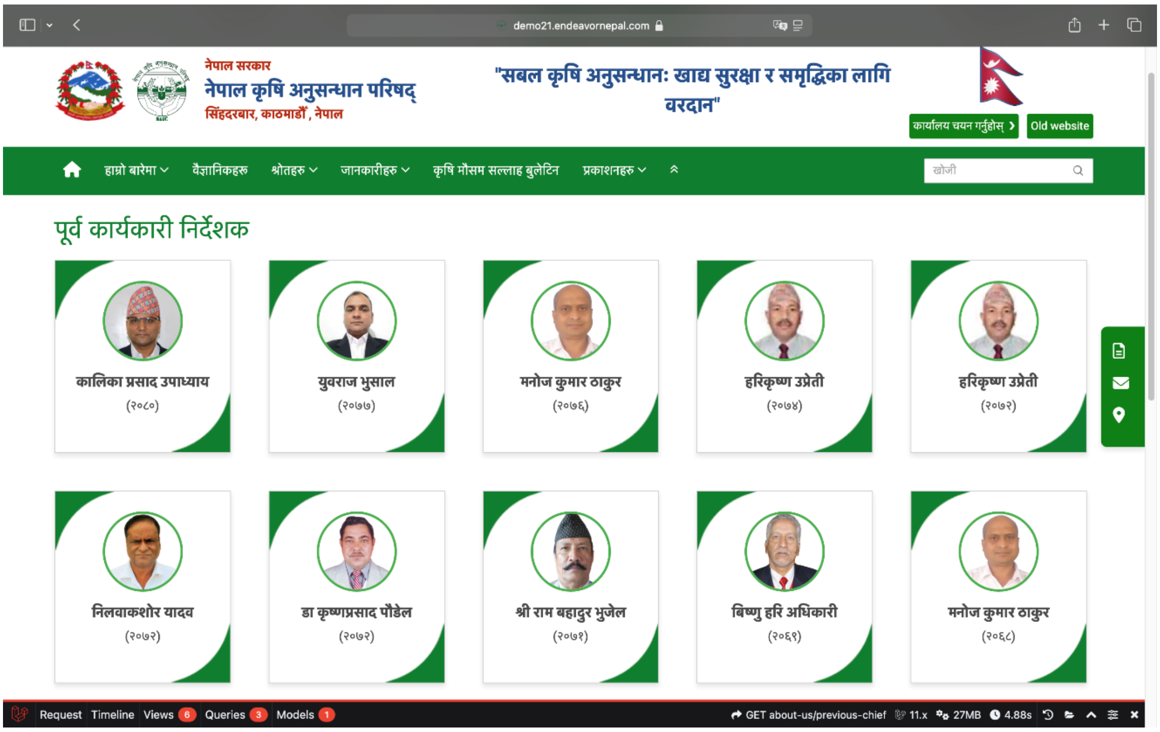
Link management index



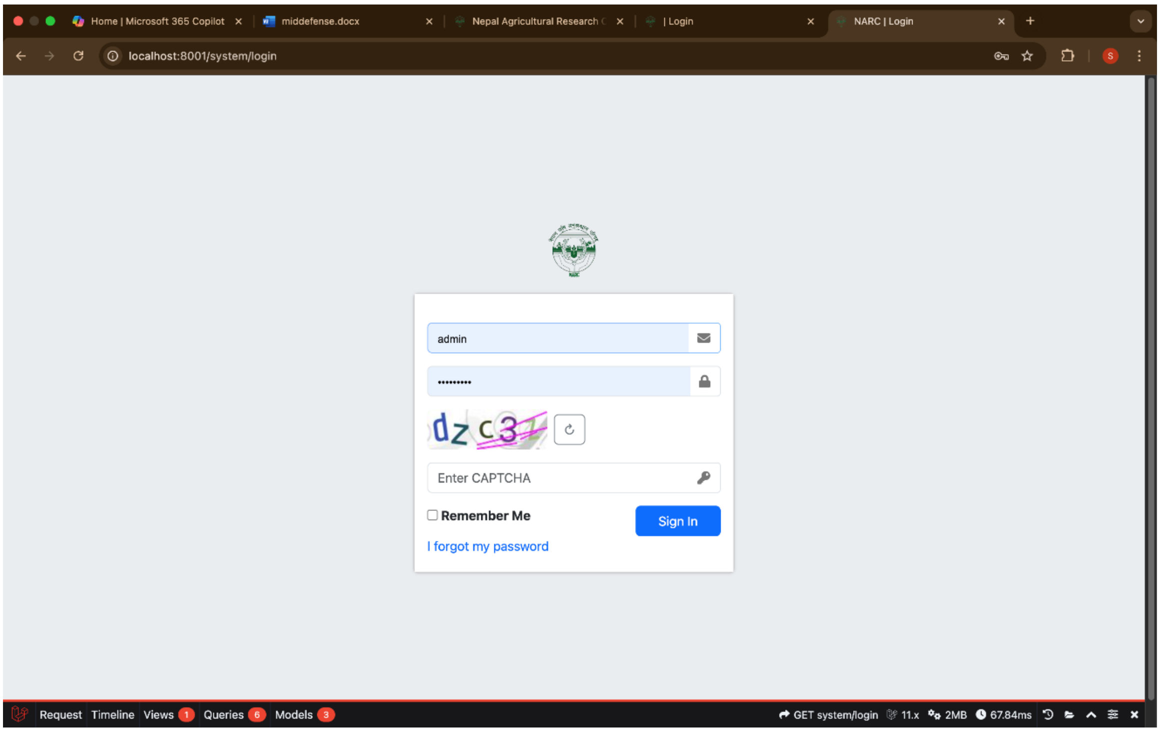
Link Management Form



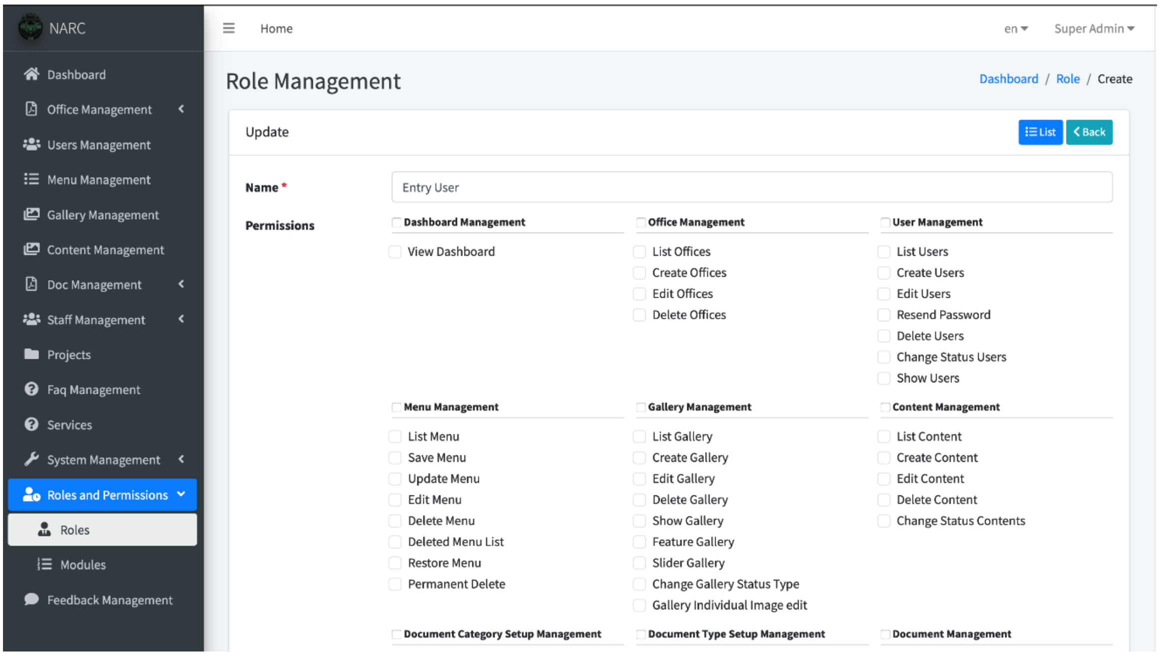
Staff Management index



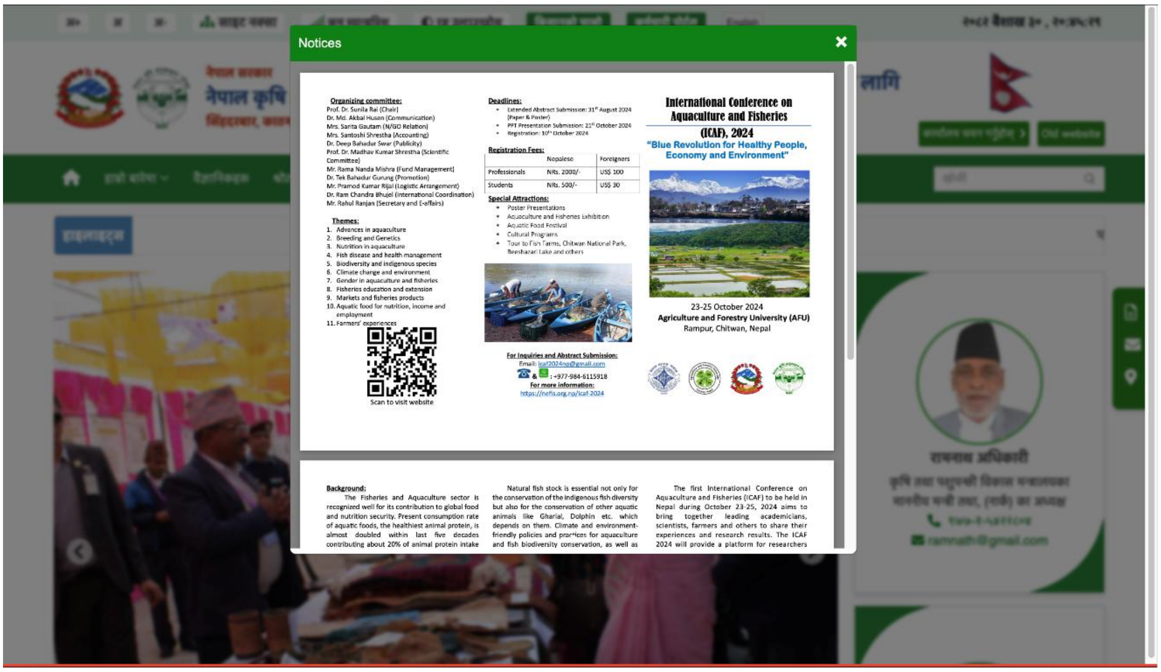
Staff list at Frontend



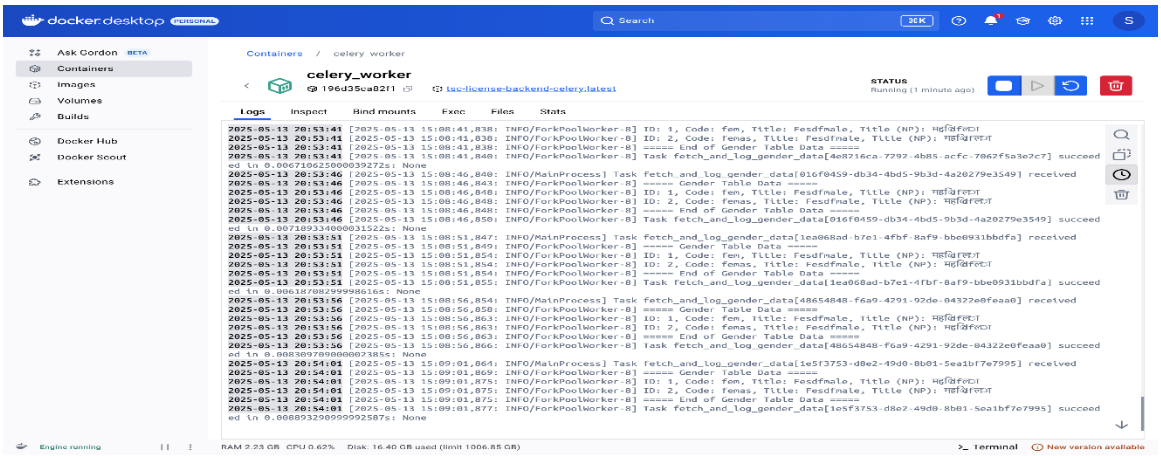
Custom Captcha at Login page

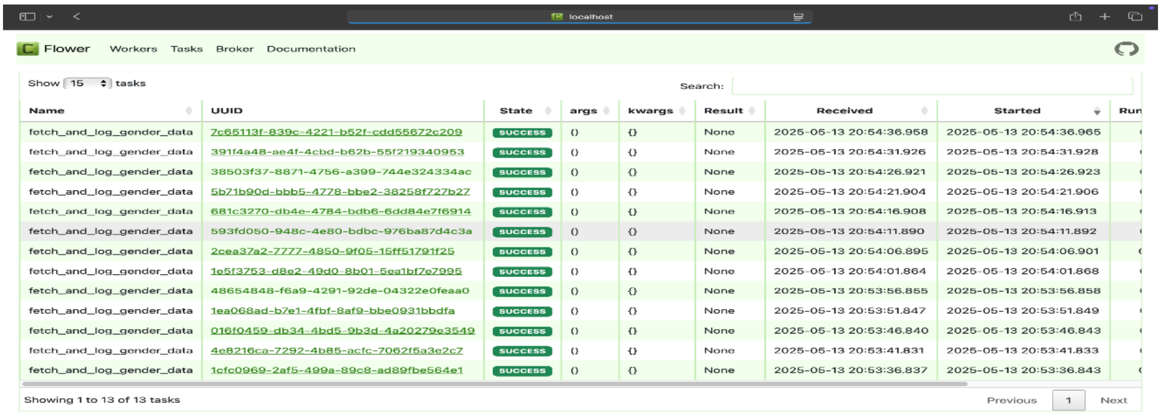


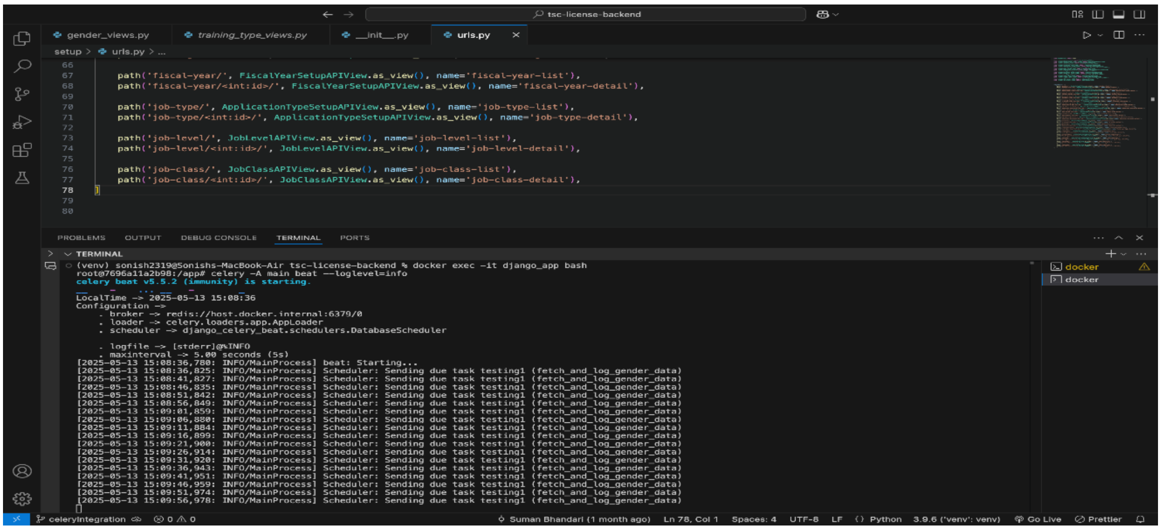
Role and permission management



Pop up at Home page







Celery, Celery-beat and Flower